

UNIVERSITI TEKNOLOGI MARA

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

CSC584

ENTERPRISE PROGRAMMING

GROUP: CS2404C

PREPARED BY:

NAME	STUDENT ID
NUR AIN NAJIHAH BINTI MOHD HOOD	2026116921

PREPARED FOR:

MUHAMAD RIDHWAN MOHAMAD RAZALI

TABLE OF CONTENTS

1. IMPLEMENTATION..... 2

1.1: HTML FORM..... 2

1.1.1 Home Page (index.html)..... 2

1.2 JSP..... 2

1.2.1 Add Student Profile..... 2

1.2.2 Edit Student Profile..... 4

1.2.3 List Student Profile..... 5

1.3 SERVLET PROCESSING..... 7

1.3.1 Add Student Profile..... 7

1.3.2 Edit Student Profile..... 8

1.3.3 Student Profile Item (com.profile.model)- Java Servet Class..... 10

2. OUTPUT SCREENSHOTS..... 12

2.1 HOME PAGE..... 12

2.2 ADD PROFILE FORM..... 12

2.3 PROFILE LIST..... 13

2.4 EDIT PROFILE FORM..... 13

2.5 DUPLICATE ID ALERT..... 14

3. GITHUB REPOSITORY..... 15

1. IMPLEMENTATION

1.1: HTML FORM

1.1.1 Home Page (index.html)

```html
<!DOCTYPE html>

<!--
To change this license header, choose License Headers in Project Properties.
To change this template file, choose Tools | Templates
and open the template in the editor.
-->
<html>
    <head>
        <title>Profile Library Home Page</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <link rel="stylesheet" href="css/bootstrap.min.css">

    </head>
    <body>
        <div class="container mt-5">
            <h1 class="mb-3">Welcome to the Student Profile Library Application</h1>
            <p class="lead">Use the links below to manage student profile:</p>

            <div class="d-flex gap-2">
                <a href="addprofile.jsp" class="btn btn-primary">Add Student Profile</a>
                <a href="listprofile.jsp" class="btn btn-secondary">View Student Profile Library</a>
            </div>
        </div>
    </body>
</html>
```

1.2 JSP

1.2.1 Add Student Profile

- Code gets the profile list from the session and builds a text of all existing student IDs.
- The JavaScript later uses it to stop the user from adding a duplicate student ID.

```jsp
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.List, com.profile.model.ProfileItem"%>

<%-- ===== JAVA: prepare the list of existing IDs ===== --%>
<%
    List<ProfileItem> profiles = (List<ProfileItem>) session.getAttribute("profileList")

       StringBuilder ids = new StringBuilder();
    if (profiles != null) {
        for (ProfileItem p : profiles) {
            String safeId = p.getId().replace("\\", "\\\\").replace("\"", "\\\"");
            if (ids.length() > 0) {
                ids.append(",");
            }
            ids.append("\"").append(safeId).append("\"");
        }
    }
%>
```

- The <head> links the Bootstrap CSS file and stores the existing student IDs in a JavaScript array.
- The checkForm() function compares the new ID with that array and shows an alert to stop a duplicate.

```
<!DOCTYPE html>
<html>
    <head>
        <title>Add Student Profile</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <link rel="stylesheet" href="css/bootstrap.min.css">

        <%-- ===== JAVASCRIPT: block duplicate student IDs ===== --%>
        <script>
            var existingIds = [<%= ids %>];

            function checkForm() {
                var newId = document.getElementById("studentId").value.trim().toLowerCase();

                for (var i = 0; i < existingIds.length; i++) {
                    if (existingIds[i].trim().toLowerCase() === newId) {
                        alert("Student ID already exists. Please enter a different ID.");
                        return false;   // stops the form from submitting
                    }
                }
                return true;            // allows the submit
            }
        </script>
```

- The form sends the data to AddProfileServlet using method="post", and runs checkForm() before it submits.
- Each field has a label and a name that the servlet reads, styled with Bootstrap classes.

```
    </head>


        <body>
        <div class="container mt-4">
            <h1 class="mb-4">Add Student Profile</h1>

            <form action="AddProfileServlet" method="post" onsubmit="return checkForm()">

                <div class="mb-3">
                    <label for="name" class="form-label">Student Name</label>
                    <input type="text" class="form-control" id="name" name="name" required>
                </div>

                <div class="mb-3">
                    <label for="studentId" class="form-label">Student ID</label>
                    <input type="text" class="form-control" id="studentId" name="id" required>
                </div>

                <div class="mb-3">
                    <label for="programme" class="form-label">Programme</label>
                    <input type="text" class="form-control" id="programme" name="programme" required>
                </div>

                <div class="mb-3">
                    <label for="email" class="form-label">Email</label>
                    <input type="email" class="form-control" id="email" name="email" required>
                </div>

                <div class="mb-3">
                    <label for="hobbies" class="form-label">Hobbies</label>
                    <input type="text" class="form-control" id="hobbies" name="hobbies" required>
                </div>

                <div class="mb-3">
                    <label for="selfintro" class="form-label">Short Self-introduction</label>
                    <input type="text" class="form-control" id="selfintro" name="selfintro" required>
                </div>

                <input type="submit" value="Add Profile" class="btn btn-primary">
                <a href="listprofile.jsp" class="btn btn-secondary">Back to list</a>
            </form>
        </div>
    </body>
</html>
```

1.2.2 Edit Student Profile

- **JSP (Java):** the scriptlet reads the `index` from the URL and gets that profile from the session list, then stores its details in variables to fill in the form.
- **JSP (Java):** it also builds a text of all existing student IDs for the duplicate check.

```jsp
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.List, com.profile.model.ProfileItem"%>

<%-- ===== JAVA: load the profile and prepare the existing IDs ===== --%>
<%
    //find profile to be edited
    int index = Integer.parseInt(request.getParameter("index"));

    // retrieve profile
    List<ProfileItem> profiles = (List<ProfileItem>) session.getAttribute("profileList");
    ProfileItem p = profiles.get(index);

    String name      = p.getName();
    String id        = p.getId();
    String programme = p.getProgramme();
    String email     = p.getEmail();
    String hobbies   = p.getHobbies();
    String selfintro = p.getSelfintro();

    StringBuilder ids = new StringBuilder();
    for (ProfileItem item : profiles) {
        String safeId = item.getId().replace("\\", "\\\\").replace("\"", "\\\"");
        if (ids.length() > 0) {
            ids.append(",");
        }
        ids.append("\"").append(safeId).append("\"");
    }
%>
```

- **JSP:** <%= index %> and <%= ids %> put the current profile position and the existing IDs into the page.
- **JavaScript:** checkForm() compares the new ID with the existing IDs and shows an alert to stop a duplicate. It skips the profile being edited, so the user can save without changing the ID.

```jsp
<!DOCTYPE html>
<html>
    <head>
        <title>Edit Student Profile</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <link rel="stylesheet" href="css/bootstrap.min.css">

        <%-- javascript to check if id duplicate --%>
        <script>
            var currentIndex = <%= index %>;
            var existingIds = [<%= ids %>];

            function checkForm() {
                var newId = document.getElementById("studentId").value.trim().toLowerCase();

                for (var i = 0; i < existingIds.length; i++) {
                    if (i === currentIndex) {
                        continue;   // skip the profile being edited
                    }
                    if (existingIds[i].trim().toLowerCase() === newId) {
                        alert("Student ID already exists. Please enter a different ID.");
                        return false;   // stops the form from submitting
                    }
                }
                return true;            // allows the submit
            }
        </script>
    </head>
```

- **HTML:** the form sends the data to EditProfileServlet using method="post", and a hidden input sends the profile index so the servlet knows which profile to update.
- **JSP:** <%= name %>, <%= id %> and the other tags fill each input with the current details.
- **Bootstrap classes:** container, form-control and btn style the layout.
- **JavaScript:** onsubmit="return checkForm()" runs the duplicate check before the form submits.

```
    <body>
        <div class="container mt-4">
            <h1 class="mb-4">Edit Student Profile</h1>

            <form action="EditProfileServlet" method="post" onsubmit="return checkForm()">
                <input type="hidden" name="index" value="<%= index %>">

                <div class="mb-3">
                    <label for="name" class="form-label">Student Name</label>
                    <input type="text" class="form-control" id="name" name="name" value="<%= name %>" required>
                </div>

                <div class="mb-3">
                    <label for="studentId" class="form-label">Student ID</label>
                    <input type="text" class="form-control" id="studentId" name="id" value="<%= id %>" required>
                </div>

                <div class="mb-3">
                    <label for="programme" class="form-label">Programme</label>
                    <input type="text" class="form-control" id="programme" name="programme" value="<%= programme %>" required>
                </div>

                <div class="mb-3">
                    <label for="email" class="form-label">Email</label>
                    <input type="email" class="form-control" id="email" name="email" value="<%= email %>" required>
                </div>

                <div class="mb-3">
                    <label for="hobbies" class="form-label">Hobbies</label>
                    <input type="text" class="form-control" id="hobbies" name="hobbies" value="<%= hobbies %>" required>
                </div>

                <div class="mb-3">
                    <label for="selfintro" class="form-label">Short Self-introduction</label>
                    <input type="text" class="form-control" id="selfintro" name="selfintro" value="<%= selfintro %>" required>
                </div>

                <input type="submit" value="Save Changes" class="btn btn-primary">
                <a href="listprofile.jsp" class="btn btn-secondary">Back to list</a>
            </form>
        </div>
    </body>
</html>
```

1.2.3 List Student Profile

- **HTML:** the <head> links the Bootstrap CSS file. A small <style> turns the border width back on, because the Vapor theme sets it to 0.
- **JSP:** the scriptlet gets the message from the request and shows it in a green Bootstrap alert if there is one. It then gets the profile list from the session.

```jsp
<%@page import="java.util.List"%>
<%@page import="com.profile.model.ProfileItem"%>
<%@page import="java.sql.*" %>

<!DOCTYPE html>
<html>
<head>
    <title>Student Profile Library</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="css/bootstrap.min.css">
    <style>
        :root { --bs-border-width: 1px; }
    </style>
</head>
<body>
    <div class="container mt-4">
        <h1 class="mb-4">Student Profile Library</h1>

        <!-- Display a message if there was any (e.g., after adding new music) -->
        <%
            String message = (String) request.getAttribute("message");
            if (message != null) {
        %>
            <div class="alert alert-success"><%= message %></div>
        <%

            }
                List<ProfileItem> profiles = (List<ProfileItem>) session.getAttribute("profileList");
        %>
```

- **HTML:** the <table> has a header row with the column names.
- **JSP (Java):** if the list is empty, it shows “No profiles added yet.” Otherwise, a for loop goes through each profile and prints one row. <%= p.getName() %> and the other tags fill in the details.
- **JSP:** each row has an Edit button that links to editprofile.jsp?index=<%= i %>, so the edit page knows which profile to open.
- **Bootstrap classes:** table, table-bordered, table-hover and table-responsive style the table.

```jsp

        <div class="table-responsive">
            <table class="table table-bordered table-hover align-middle">
                <thead>
                    <tr>
                        <th>Number</th>
                        <th>Student Name</th>
                        <th>Student ID</th>
                        <th>Programme</th>
                        <th>Email</th>
                        <th>Hobbies</th>
                        <th>Short Self-introduction</th>
                        <th>Action</th>
                    </tr>
                </thead>
                <tbody>
                    <%
                        if (profiles == null || profiles.isEmpty()) {
                    %>
                    <tr>
                        <td colspan="8" class="text-center">No profiles added yet.</td>
                    </tr>
                    <%
                        } else {
                            for (int i = 0; i < profiles.size(); i++) {
                                ProfileItem p = profiles.get(i);
                    %>
                    <tr>
                        <td><%= i + 1 %></td>
                        <td><%= p.getName() %></td>
                        <td><%= p.getId() %></td>
                        <td><%= p.getProgramme() %></td>
                        <td><%= p.getEmail() %></td>
                        <td><%= p.getHobbies() %></td>
                        <td><%= p.getSelfintro() %></td>
                        <td><a href="editprofile.jsp?index=<%= i %>" class="btn btn-sm btn-primary">Edit</a></td>
                    </tr>
                    <%
                            }
                        }
                    %>
                </tbody>
            </table>
        </div>

```

- **HTML:** the “Add New Profile” button links to addprofile.jsp.
- **Bootstrap classes:** btn btn-secondary styles the button, and the remaining tags close the page.

```
            <a href="addprofile.jsp" class="btn btn-secondary">Add New Profile</a>
        </div>
    </body>
    </html>
```

1.3 SERVLET PROCESSING

1.3.1 Add Student Profile

- **Java:** the imports load the servlet classes, HttpSession, List and ProfileItem.
- **Java Servlet:** processRequest reads the six form values (name, id, programme, email, hobbies, selfintro) from the request using request.getParameter(). These names match the name attributes in addprofile.jsp.
- **Java Servlet:** it gets the profile list from the session, or creates a new list if there is none, because the app has no database. It then adds a new ProfileItem with the submitted details.
- **Java Servlet:** it ends with sendRedirect("listprofile.jsp"), which sends the user to the list page to see the new profile.

```
import com.profile.model.ProfileItem;
import java.io.IOException;
import java.util.ArrayList;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 *
 * @author USER
 */
public class AddProfileServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

        String name= request.getParameter("name");
        String id= request.getParameter("id");
        String programme = request.getParameter("programme");
        String email = request.getParameter("email");
        String hobbies = request.getParameter("hobbies");
        String selfintro = request.getParameter("selfintro");

        // No database: keep the profiles in the session
        HttpSession session = request.getSession();
        List<ProfileItem> profiles = (List<ProfileItem>) session.getAttribute("profileList");
        if (profiles == null) {
            profiles = new ArrayList<>();
            session.setAttribute("profileList", profiles);
        }
        profiles.add(new ProfileItem(name, id, programme, email, hobbies, selfintro));

        response.sendRedirect("listprofile.jsp");
    }
```

- **Java Servlet:** doPost runs when the form is submitted with method="post", and it calls processRequest to save the profile.
- **Java Servlet:** doGet runs when someone opens the servlet link directly, and it just redirects to addprofile.jsp instead of saving anything.

```
39
40          // A direct visit or link click (GET) just opens the form
41          @Override
            protected void doGet(HttpServletRequest request, HttpServletResponse response)
43                  throws ServletException, IOException {
44              response.sendRedirect("addprofile.jsp");
45          }
46
47          // The form submit (POST) does the insert
48          @Override
            protected void doPost(HttpServletRequest request, HttpServletResponse response)
50                  throws ServletException, IOException {
51              processRequest(request, response);
52          }
53
54      }
```

1.3.2 Edit Student Profile

- **Java:** the imports load the servlet classes, HttpSession, List and ProfileItem.
- **Java Servlet:** processRequest reads the six form values (name, id, programme, email, hobbies, selfintro) and the hidden index from the request using request.getParameter(). These names match the name attributes in editprofile.jsp.
- **Java Servlet:** it gets the profile list from the session, then uses the index to find the profile being edited. It checks that the index is valid before updating.
- **Java Servlet:** it replaces the old details with the new ones using the set methods of ProfileItem. The try/catch handles a missing or invalid index, so the page does not crash.
- **Java Servlet:** it ends with sendRedirect("listprofile.jsp"), which sends the user back to the list to see the updated profile.

```java
import com.profile.model.ProfileItem;
import java.io.IOException;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class EditProfileServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

        HttpSession session = request.getSession(false);


        //RETRIEVE
        String name= request.getParameter("name");
        String id= request.getParameter("id");
        String programme = request.getParameter("programme");
        String email = request.getParameter("email");
        String hobbies = request.getParameter("hobbies");
        String selfintro = request.getParameter("selfintro");

        List<ProfileItem> profiles = (List<ProfileItem>) session.getAttribute("profileList");

        String indexStr = request.getParameter("index");
        try {
            int index = Integer.parseInt(indexStr);
            if (profiles != null && index >= 0 && index < profiles.size()) {
                // UPDATE (replaces the SQL UPDATE)
                ProfileItem p = profiles.get(index);
                p.setName(name);
                p.setId(id);
                p.setProgramme(programme);
                p.setEmail(email);
                p.setHobbies(hobbies);
                p.setSelfintro(selfintro);
            }
        } catch (NumberFormatException e) {
            // missing or invalid index: nothing to update
        }

        response.sendRedirect("listprofile.jsp");
    }
```

- **Java Servlet:** doPost runs when the edit form is submitted with method="post", and it calls processRequest to save the changes.
- **Java Servlet:** doGet runs when someone opens the servlet link directly, and it redirects instead of changing any data.

```java
    // A direct visit or link click (GET) just opens the form
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.sendRedirect("listprofile.jsp");
    }

    // The form submit (POST) does the insert
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

}
```

1.3.3 Student Profile Item (com.profile.model) - Java Servlet Class

- Constructor

```java
package com.profile.model;

public class ProfileItem implements java.io.Serializable {
    private String name;
    private String id;
    private String programme;
    private String email;
    private String hobbies;
    private String selfintro;


    //constructor
    public ProfileItem()
    {
        name = "";
        id = "";
        programme = "";
        email = "";
        hobbies = "";
        selfintro = "";
    }

    public ProfileItem(String name, String id, String programme, String email, String hobbies, String selfintro)
    {
        this.name = name;
        this.id = id;
        this.programme = programme;
        this.email = email;
        this.hobbies = hobbies;
        this.selfintro = selfintro;
    }
```

- Getter

```java
    //getter
    public String getName()
    {
        return name;
    }

    public String getId ()
    {
        return id;
    }

    public String getProgramme ()
    {
        return programme;
    }

    public String getEmail ()
    {
        return email;
    }

    public String getHobbies ()
    {
        return hobbies;
    }

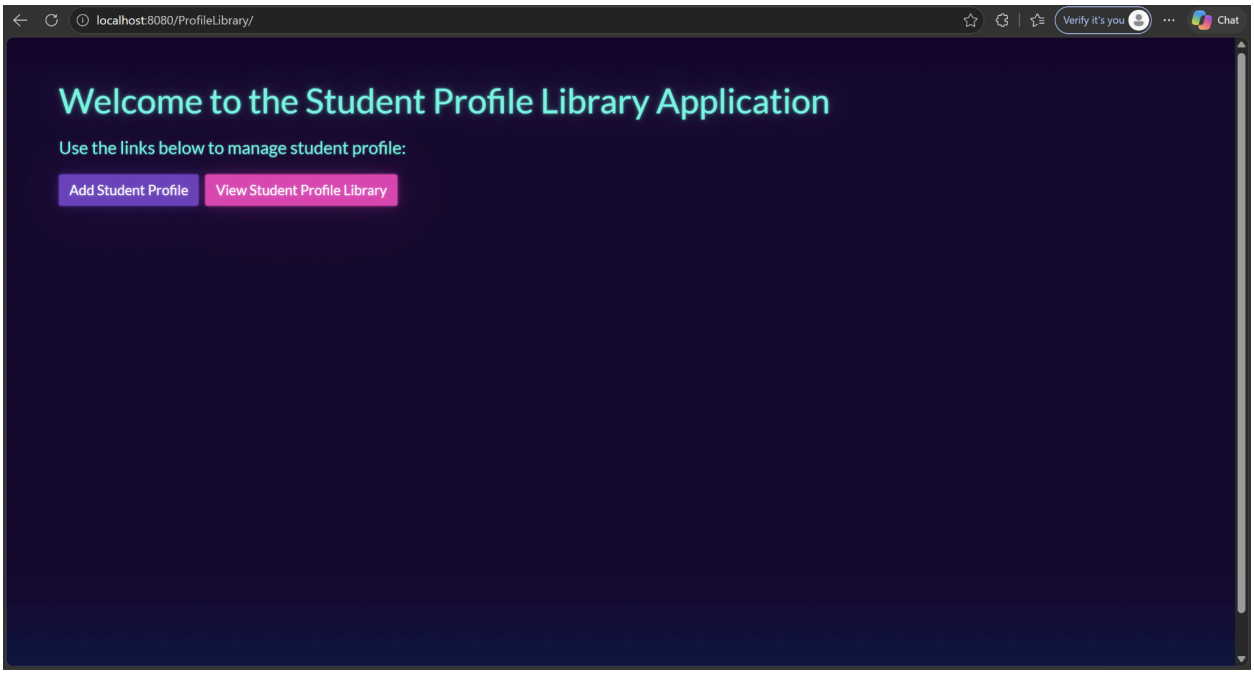
    public String getSelfintro ()
    {
        return selfintro;
    }
```

- Setter

```
66      //setter
67      public void setName (String name){
68          this.name = name;
69      }
70
71      public void setId (String id){
72          this.id = id;
73      }
74
75      public void setProgramme (String programme){
76          this.programme = programme;
77      }
78
79      public void setEmail (String email){
80          this.email = email;
81      }
82
83      public void setHobbies (String hobbies){
84          this.hobbies = hobbies;
85      }
86
87      public void setSelfintro(String selfintro){
88          this.selfintro = selfintro;
89      }
90
91
92
93  }
```

2. OUTPUT SCREENSHOTS

2.1 HOME PAGE



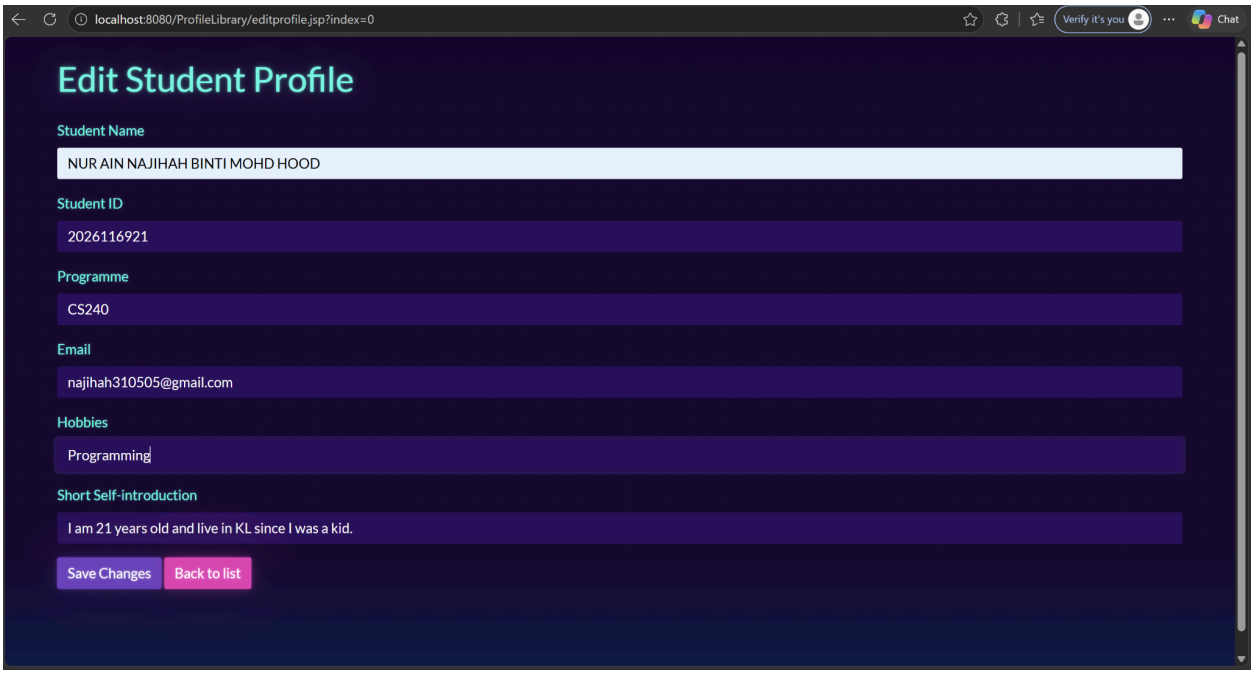
2.2 ADD STUDENT PROFILE FORM



2.3 STUDENT PROFILE LIST



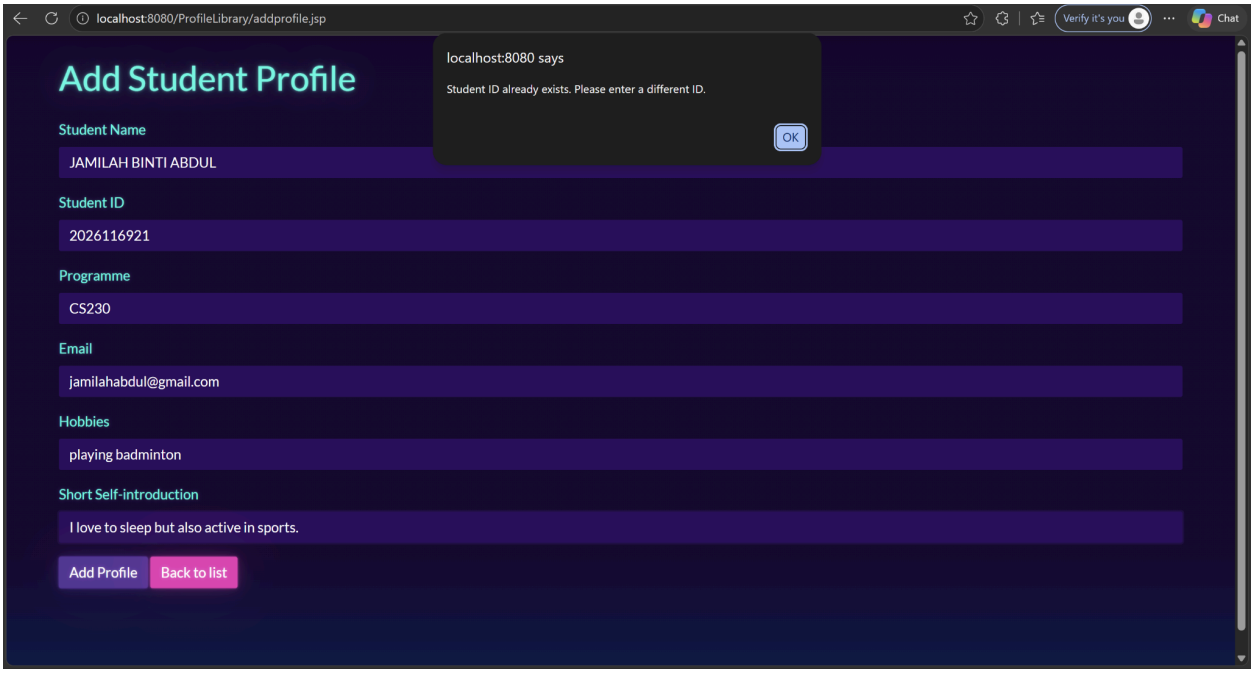
2.4 EDIT STUDENT PROFILE FORM



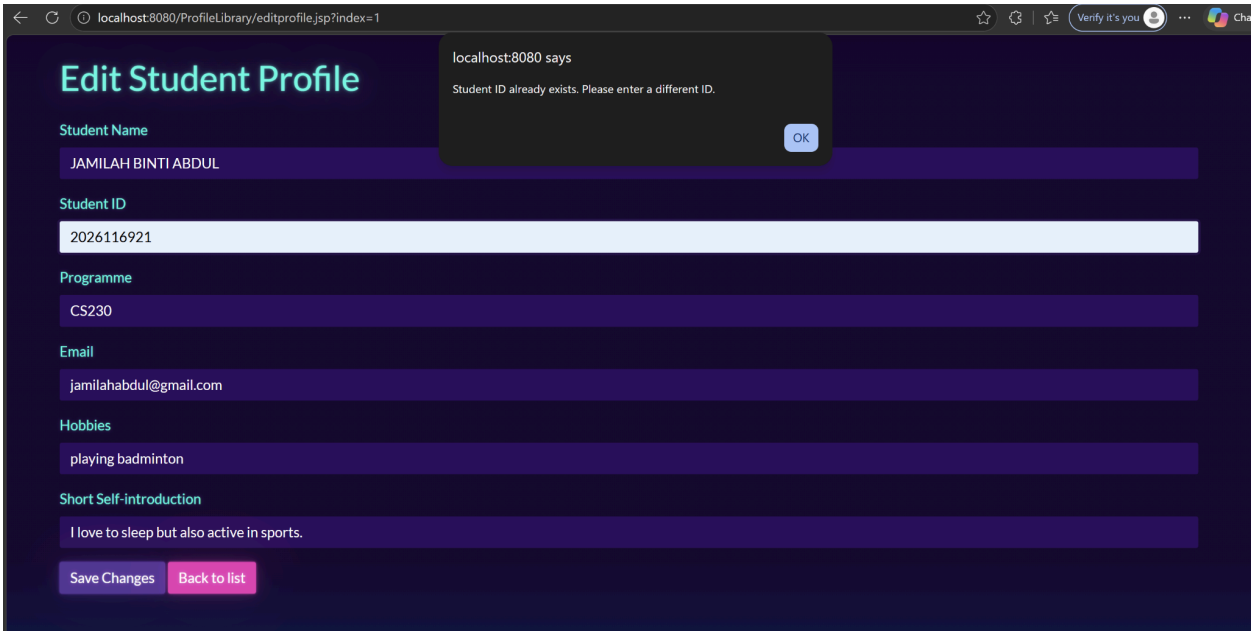


2.5 DUPLICATE STUDENT ID ALERT

On addprofile.jsp:



On editprofile.jsp:



3. GITHUB REPOSITORY

https://github.com/tantantara/CSC584-Individual-Assignment--1